Graph reconstruction via subgraph queries

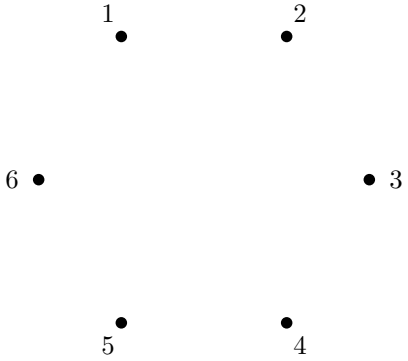
Bianca Oancea

LISN, Université Paris-Saclay

JITA, October 2026

Discovering a labelled graph

Discover which of the 15 edges are there!

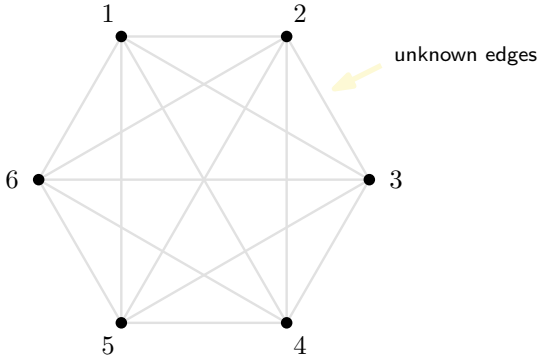


$n = 6,\ k = 3$

$\binom{6}{3}$ sets of vertices we can ask about

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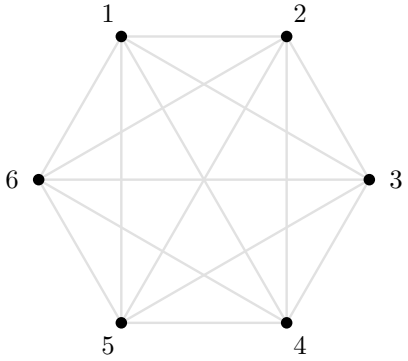
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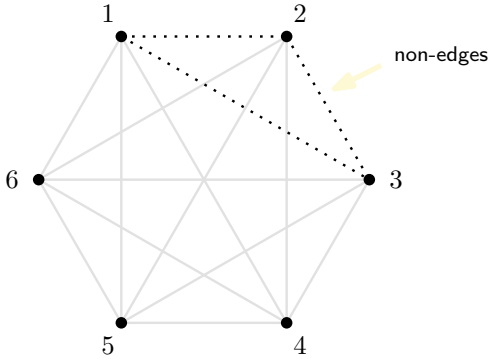
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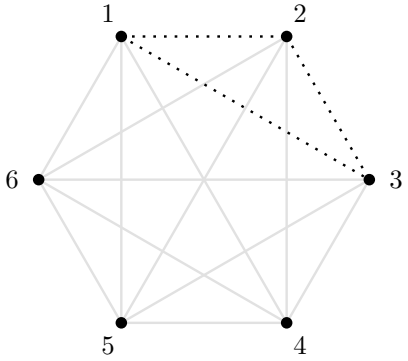
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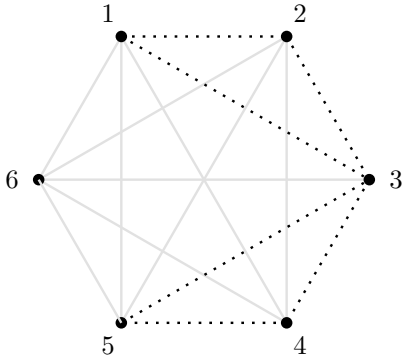
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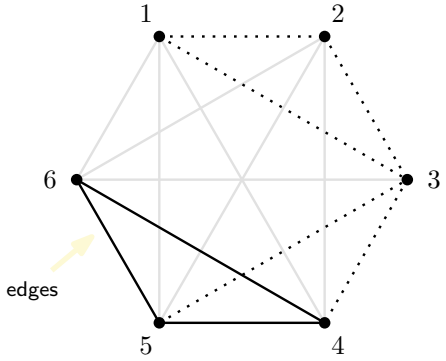
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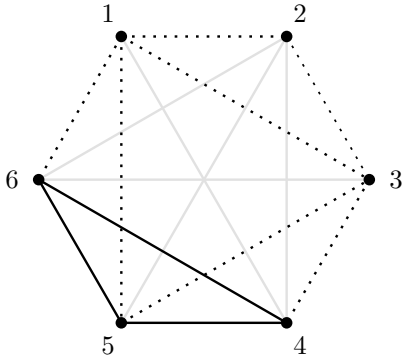


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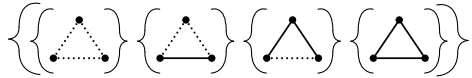
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Reconstruction via subgraph queries

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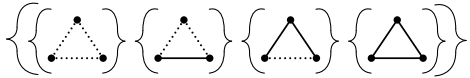
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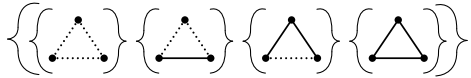
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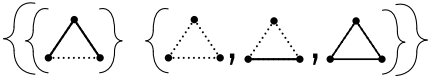
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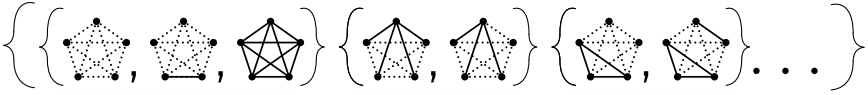
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Queries on P_3- structures (Hayward, 1996)

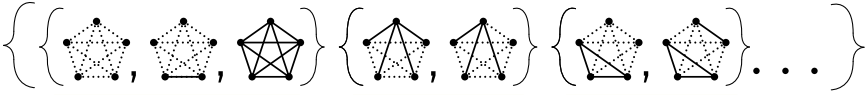


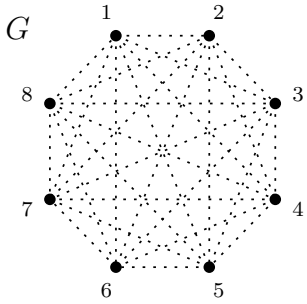
Which partitions give enough information?

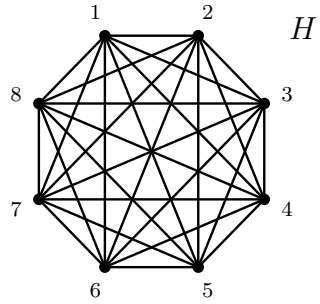
$\Pi =$ 



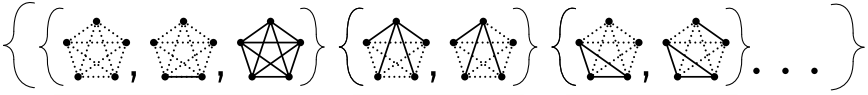
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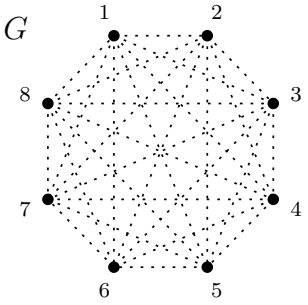


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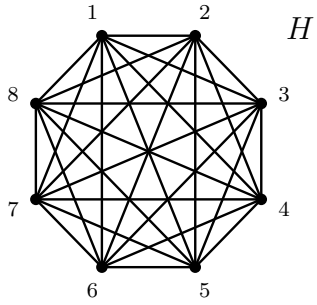
A B C

G



$\Pi(G[12345]) = A$

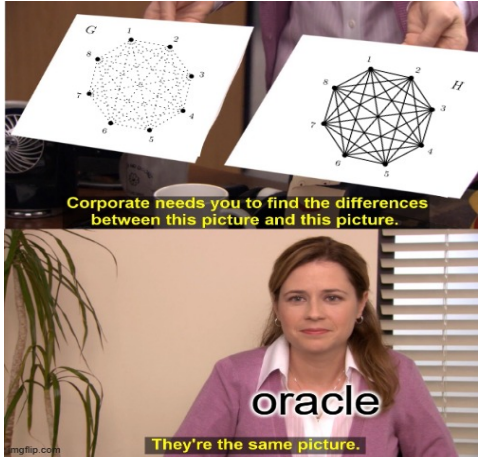
H



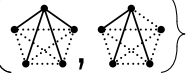
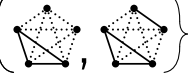
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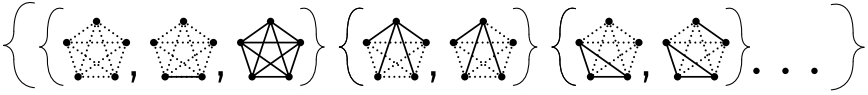
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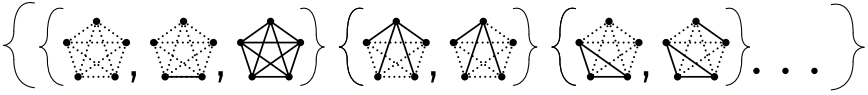
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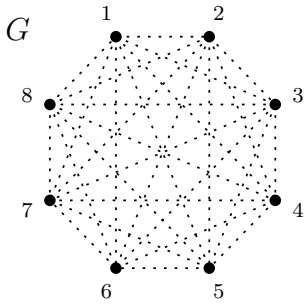
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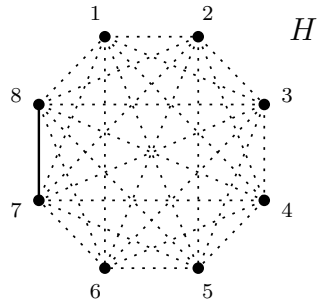
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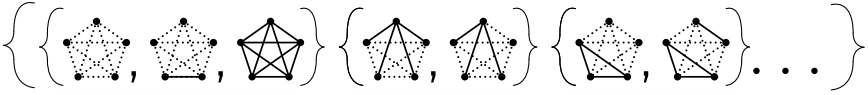
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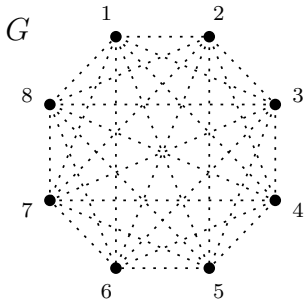
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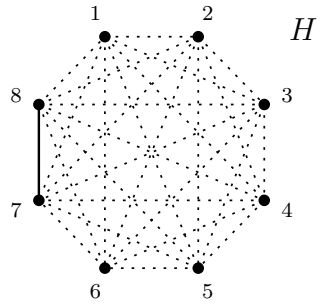
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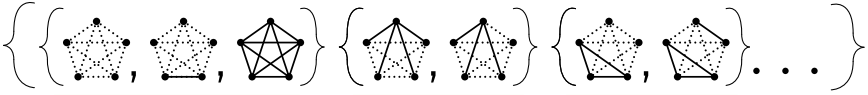
$\vdots$



$\Pi(G[12345]) = A$

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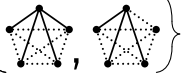
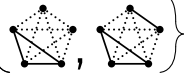
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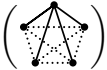
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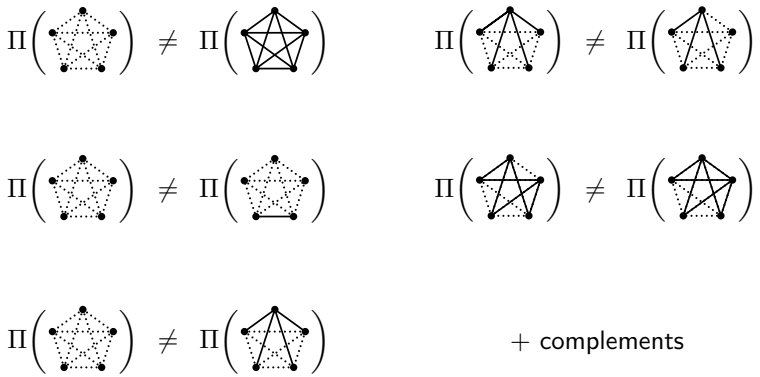
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Cohen, La, O., 2026+

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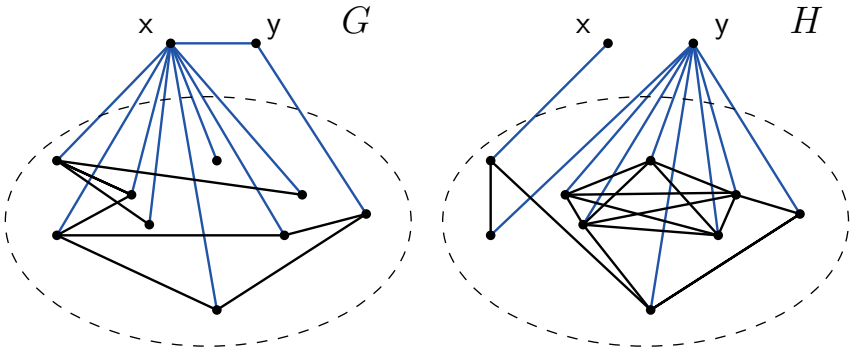
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→ the nine separations correspond to $r = 2$, there is an explicit family of separations for every $r \geq 3$.

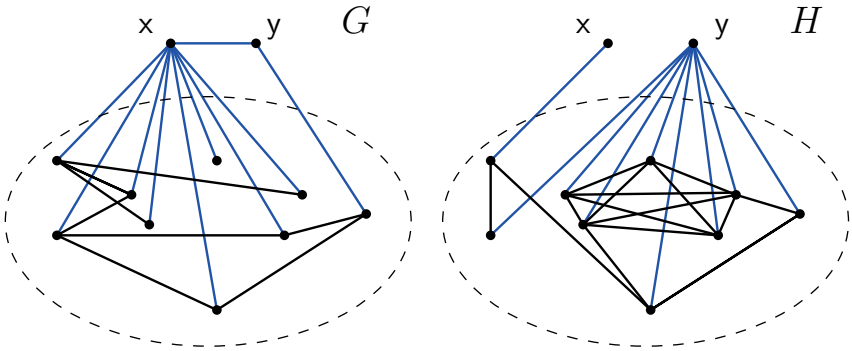
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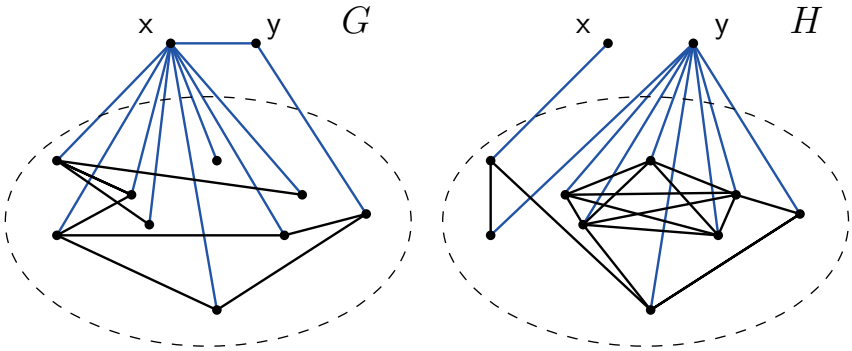
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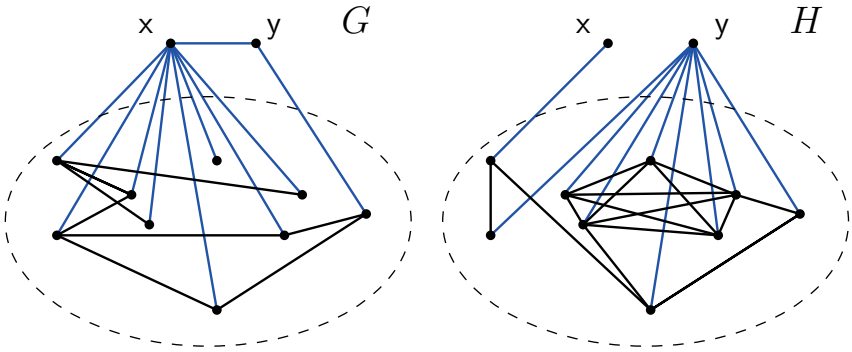


Step 1: find a subset U, such that x, y is adjacent to all or none of U, in each graph

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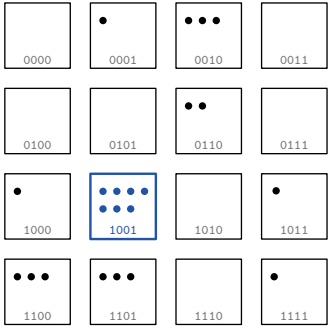


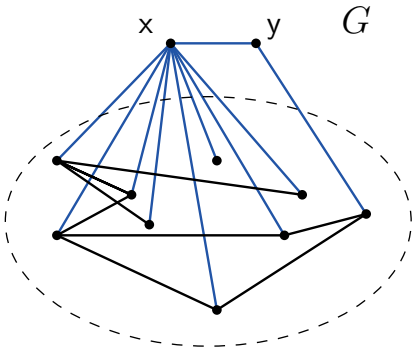
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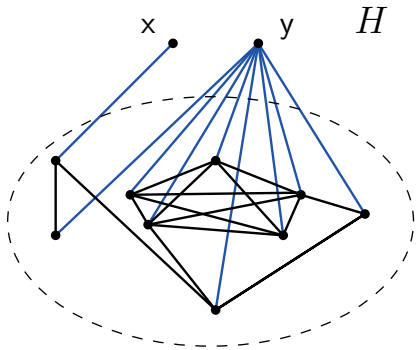
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pigeonhole principle



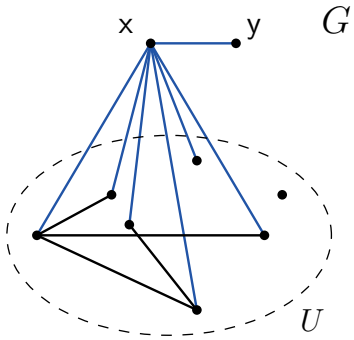


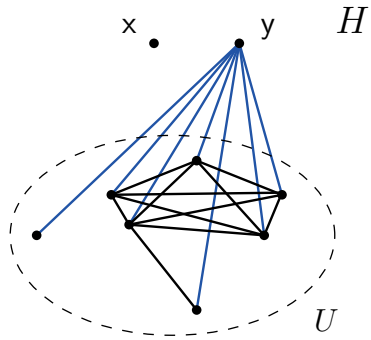


Step 1: find a subset U, such that x, y is adjacent to all or none of U, in each graph

Sketch of proof

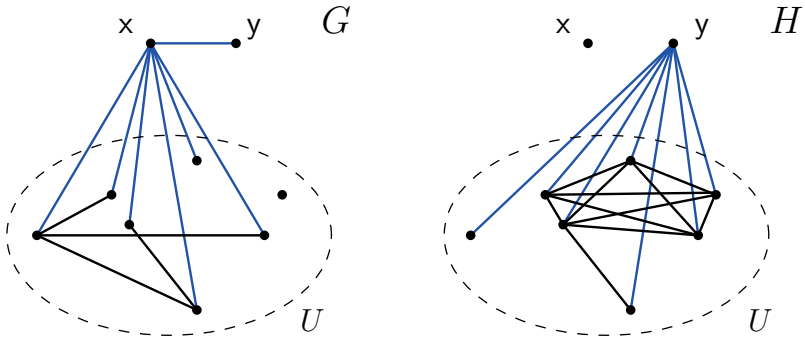
Goal: find S, $|S| = k$, with $\Pi(G[S]) \neq \Pi(H[S])$





Sketch of proof

Goal: find S, $|S| = k$, with $\Pi(G[S]) \neq \Pi(H[S])$



Step 2: find a subset $D \subseteq U$, $|D| \geq k$ such that D induces a complete or empty graph in each graph, G and H

Sketch of proof

Goal: find S, $|S| = k$, with $\Pi(G[S]) \neq \Pi(H[S])$

multicolor Ramsey theorem

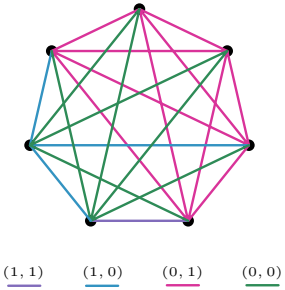
For all integers $c \geq 1$ and $k \geq 2$, there is an integer N such that every coloring of the pairs of an N-element set with c colors has a set of size k whose pairs all have the same color.

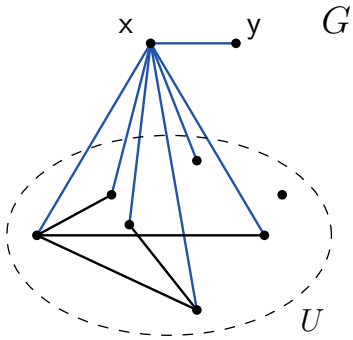
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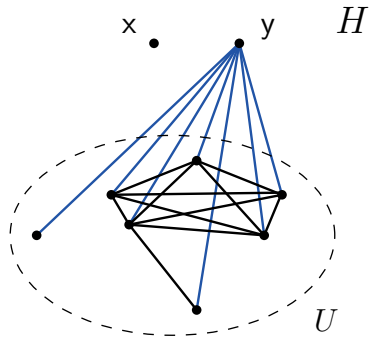
Sketch of proof

Goal: find S, $|S| = k$, with $\Pi(G[S]) \neq \Pi(H[S])$

four-color Ramsey theorem



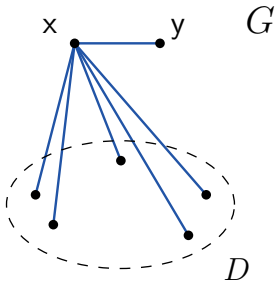


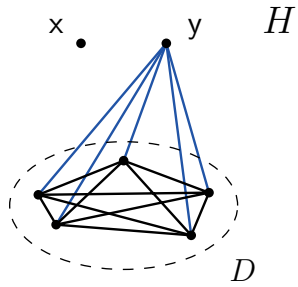


Step 2: find a subset $D \subseteq U$, $|D| \geq k$ such that D induces a complete or empty graph in each graph, G and H

Sketch of proof

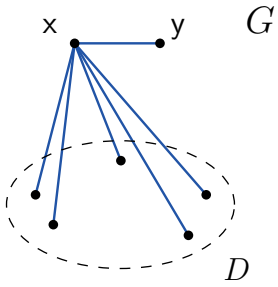
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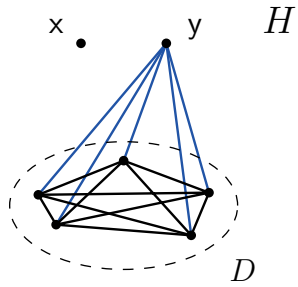




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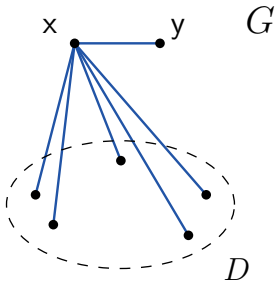


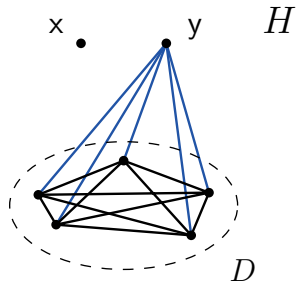
Step 3: represent all edges on $D \cup \{x, y\}$ with four bits (D, xD, yD, xy)

Sketch of proof

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Pattern:
(D, xD, yD, xy)



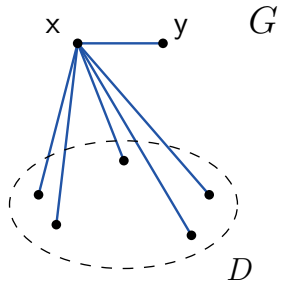


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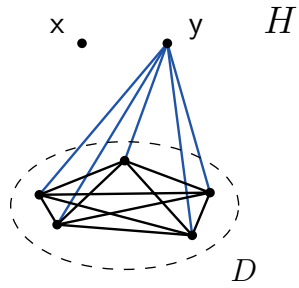
Sketch of proof

Goal: find S, $|S| = k$, with $\Pi(G[S]) \neq \Pi(H[S])$

Pattern:
(D, xD, yD, xy)



$(0, 1, 0, 1)$



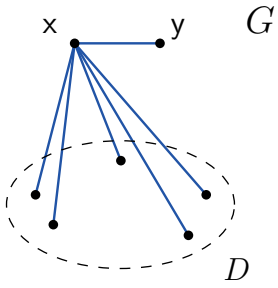
$(1, 0, 1, 0)$

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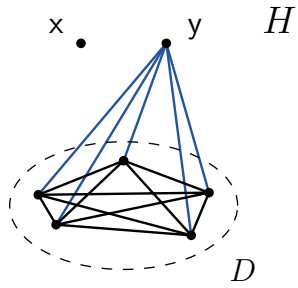
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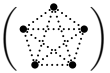
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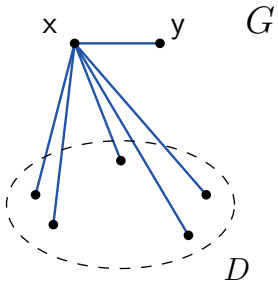
Step 4: find the first disagreement in the two patterns and choose S (exemple: k vertices from D)

Sketch of proof

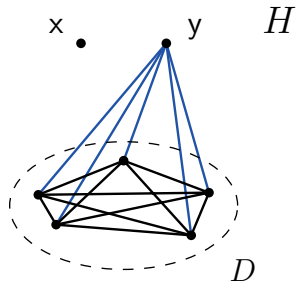
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Pattern:
(D, xD, yD, xy)

Π  $\neq$ Π 



$(0, 1, 0, 1)$



$(1, 0, 1, 0)$

Step 3: represent all edges on $D \cup \{x, y\}$ with four bits (D, xD, yD, xy)

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From edge patterns to the nine separations

Pattern:
(D, xD, yD, xy)

From edge patterns to the nine separations

Pattern:	G:	H:
$(D,\, xD,\, yD,\, xy)$	$(a,\, b,\, c,\, d)$	$(a',\, b',\, c',\, d')$

From edge patterns to the nine separations

Pattern:
(D, xD, yD, xy)

G:
(a, b, c, d)

H:
(a', b', c', d')

First difference

Which S to query

The induced graphs

From edge patterns to the nine separations

Pattern:
$(D,\, xD,\, yD,\, xy)$

G:
$(a,\, b,\, c,\, d)$

H:
$(a',\, b',\, c',\, d')$

First difference

$a \neq a'$

Which S to query

k vertices of D

The induced graphs

From edge patterns to the nine separations

Pattern:
$(D,\, xD,\, yD,\, xy)$

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$(a,\, b,\, c,\, d)$

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First difference

$a \neq a'$

Which S to query

k vertices of D

The induced graphs



From edge patterns to the nine separations

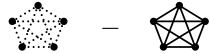
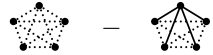
Pattern: $(D,\, xD,\, yD,\, xy)$	G: $(a,\, b,\, c,\, d)$	H: $(a',\, b',\, c',\, d')$
First difference	Which S to query	The induced graphs
$a \neq a'$	k vertices of D	
$a = a', b \neq b'$ or $a = a', c \neq c'$	x (or y) $\cup$ $k-1$ vertices of D	

From edge patterns to the nine separations

Pattern:
$(D,\ xD,\ yD,\ xy)$

G:
$(a,\ b,\ c,\ d)$

H:
$(a',\ b',\ c',\ d')$

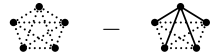
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From edge patterns to the nine separations

Pattern:
$(D,\, xD,\, yD,\, xy)$

G:
$(a,\, b,\, c,\, d)$

H:
$(a',\, b',\, c',\, d')$

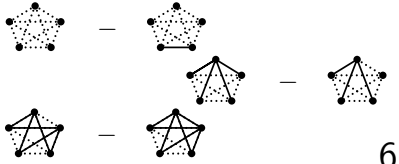
First difference	Which S to query	The induced graphs
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$a = a', b = b', c = c'$ $d \neq d'$	$\{x, y\} \cup k-2$ vertices of D	

From edge patterns to the nine separations

Pattern:
$(D,\, xD,\, yD,\, xy)$

G:
$(a,\, b,\, c,\, d)$

H:
$(a',\, b',\, c',\, d')$

First difference	Which S to query	The induced graphs
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Conclusion and open questions

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Cohen, La, O., 2026+

$N(k) = 2^{\Theta(k)}$

The partition determines every graph on $n \geq N(k)$ vertices, for any $k \geq 4$, **if and only if** it satisfies these nine separations.

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Future directions

Conclusion and open questions

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Additional conditions on the partition that lower the threshold on n

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Future directions

Additional conditions on the partition that lower the threshold on n

A more efficient reconstruction algorithm

Generalize the reconstruction theorem for directed graphs, edge-coloured graphs, posets, etc

Thank you

All nine separations

$\Pi($  $) \neq \Pi($  $)$

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